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An Empirical Review on Image Dehazing Techniques for Change Detection of Land Cover

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***Abstract*—The turbulence effects in the weather like smog, haze, and fog severely degrade the visibility and performance of outdoor images by causing texture attenuation and color decay in brightness regions, making it difficult to identify object features in the images captured or scenes. The presence of haze can affect the techniques of preprocessing, segmentation, and classification of outdoor images similar to aerial and Remote Sensing images used in the Land Use/ Land Cover (LU/ LC) change analysis. To remove atmospheric effects, dark channel prior-based and learning-based dehazing techniques have been suggested. Prior-based algorithms have the disadvantage of oversaturating the sky region, resulting in artifacts, and longer execution time. Image dehazing techniques based on image enhancement, image fusion, image reconstruction, image segmentation, and deep learning-based approaches are discussed in this review article. The review also discusses the techniques for obtaining minimum haze removal parameters and assigning appropriate values to dehazing attributes. The effectiveness of various techniques was assessed with metrics that were specific to the datasets. In the end, future research directions for image dehazing are rendered.**

Keywords*— Turbulence effects, Texture attenuation, Remote Sensing Images, Land Use Land Cover, Image Enhancement, Image Fusion, Image Segmentation, Deep learning, Dehazing attributes.

I. INTRODUCTION

The world is facing rapid, comprehensive changes and challenges in Land Use (e.g., human activities) along with Land Cover (e.g., land area's physical characteristics) [1]. The rate of population growth has many socio-economic and environmental implications including insufficient housing, urban expansion, poor transportation system, sudden pace in LU/LC, leading to the deterioration of forest and transformation of fertile soil to urban construction, all of which have a significant impact on the ecosystem [2]. According to studies, fewer landscapes in peripheral sites and remote areas on the planet are left in their natural state [3]. Changes in LU/LC reveal how human actions interact with nature [4] and are distinguished by specific factors in time and space at various levels. The list includes natural, cultural, social, political, and a variety of other influences [5].

As a result of increased research in Remote Sensing Technology (RST), aerial and remote sensing images have been used in a variety of applications, including agriculture and weather studies [6], Land Cover monitoring [7, 8], etc., With the atmospheric artifacts like haze, smog, and fog, the image scene quality is reduced and the performance of the respective application is reduced as shown in Fig. 1. The captured images lose colour and contrast details due to poor atmospheric conditions and do not provide enough information for computer vision applications.

A range of images with different atmospheric effects of the same scene is considered for the image dehazing problem [9, 10]. These methods are ineffective since acquiring several images of a similar scene in a variety of weather conditions is difficult. Later, using contrast-based and histogram-based [11, 12, 13] image enhancement techniques, several single image haze removal methods were developed.

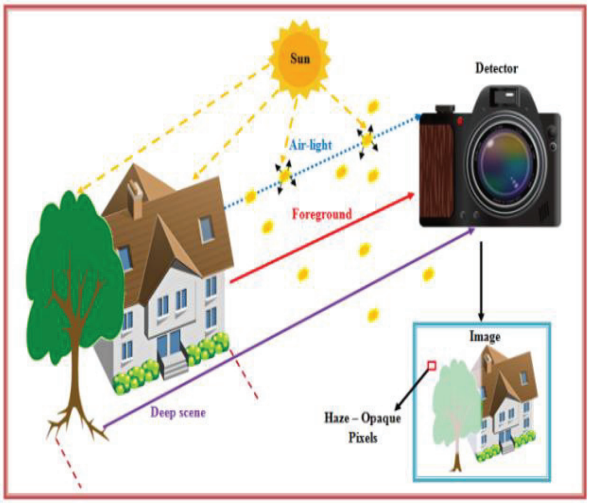

Fig. 1. Haze formation

Two main causes contributing to the haze in images are [14]:

The first reason is that the electromagnetic signal detected by optical sensors via the atmosphere layer using aircraft and satellites from the ground surface is typically subject to extreme atmospheric artifacts such as cloud, fog, smoke, haze, and many others. Since the weather is indeed not entirely free of particulate disturbances, the haziness still occurs, resulting in poor quality and efficiency.

The second reason is that man-made structures and LU/ LC particles have different surface albedos or surface reflectance coefficients.

For images degraded by the environment, many computer vision-based applications have shown deprived performance.

Objects in this type of image have lower interpretive accuracy [15] and are often seen with texture decay and strength attenuation [16]. The efficiency of computer vision applications is significantly influenced by reduced strength [17]. As a result, haze removal algorithms have the usage potential in a wide range of vision-based applications, such as object recognition and interpretation, aerial photography, and image retrieval [30]. Various haze removal techniques have indeed been extensively studied to increase the scene reliability deprived due to haze.

II. REVIEW BASED ON VARIOUS IMAGE HAZE REMOVAL TECHNIQUES

This section provides a review of various dehazing methods that can be used for aerial and satellite images. The review focuses on the various approaches and the flaws that result from them.

A. *Single Image Dehazing Methods*

Jiang *et al*. [18] introduced an observational single image haze removal technique. The obtained radiance is defined by the model as the amount of constant path global radiance, radiance reflected from an object's surface, and the spatially varied contribution of haze. This technique was developed for high and medium-resolution satellite images and it also handles scenes with varying haze depths. The results were superior in terms of efficiency, improved spatial and temporal knowledge, and spectral information consistency. However, the model created halo effects in some parts of the output image.

Sahu *et al.* [19] suggested a new colour channel in their model. In this, the hazy image was divided into 32×32 block sizes and determines the maximum score block. The chosen block was used to evaluate ambient light. Later, a new colour channel was designed to remove atmospheric scattering, ambient light, and the designed colour channel is used to evaluate the transmission depth. The radiance was then calculated using transmission depth and ambient light. The superiority of the output image is improved with intensity scaling factor. The model was evaluated using common datasets such as Outdoor-HAZE, Indoor-HAZE, FRIDA, RESIDE dataset, and other Google dataset. The downside that has been identified is an inaccurate measurement of ambient light.

Suresh Chandra et al. [20] introduced a fast and precise single image based haze removing method. The technique worked in '2' folds. To begin, a robust dehazed scaling factor was created to evaluate the precise scene depth with the haze contributed image's minimum and maximum colour channel differences. Second, to locate the image's haziest area for calculating the atmospheric light's value, a mathematical function for computing the pixel's likelihood of being at the least distance was used. The proposed method accurately recovers structural and colour details, and the computation speed has been proven to be reliable, but the contrast of the output image has been observed to be poor.

Table I illustrates the various other single image dehazing methods concerning the advantages and limitations.

TABLE I. SINGLE IMAGE DEHAZING METHODS

Author(s)	Technique(s)	Advantage(s)	Limitation(s)
Ziqi *et al.* [21]	Dense Attentive Dehazing Network (DADN)	No dependency on custom scattering model of atmosphere, enhanced performance	Unable to increase contrast in output
Tang et al. [22]	Retinex Theory and DCP Taylor series expansion	Precise estimation of transmittance, superior performance	long computational time required for dehazing
Sriparna et al. [23]	Bacterial Foraging (BF)-Fuzzy synergism	Outperforms in noise and edge detection, edge sharpening, and contrast enhancement in terms of dehazing	Method's measurement of air-light was imprecise
Zhang *et al.* [24]	DCP model focused on saliency detection with superpixel segmentation and colour contrast analysis	The transmission map was calculated with precision, there is less chromatic aberration	White regions are omitted in estimating the dark channel. As a result, ambient light prediction is
Peng *et al.* [25]	Enhanced Dark channel prior (DCP) with adaptive contrast enhancement and unsharp masking	Halo effects are no longer visible, and dehazing efficiency in areas with high concentrations of haze is enhanced	Expensive computational time, oversaturation is introduced in high colored surfaces
WangNo *et al.* [26]	Hybrid DCP combined Guided filter	Improvement in contrast and atmospheric light images, as well as customization of air-light transmission, is observed	The contrast level is distorted, and information details are lost.
Huang *et al.* [27]	Histogram and fusion processing in a multi-scale Retinex model	Preserves geometric details, and has a low degree of colour and brightness distortion after dehazing	Does not enforce images covered with wide real of sky region and oceans
Singh *et al.* [28]	Gradient Profile Prior (GPP), Guided anisotropic diffusion, and iterative learning-based image filter (GADILF)	Reduces the effect of halo and gradient artifacts by better preserving texture details, faster computational time	Spectral details and spatial resolution have not consistently improved
Wei Sun *et al.* [29]	Estimation of local ambient light using a cross bilateral filter	Errors in estimating global ambient light are greatly reduced, and halo effects are greatly reduced	If the scene depth is too high, the output can be unsatisfactory

B. *Image Enhancement Based Dehazing Methods*

Li Ping Yao *et al.* [30] devised a Retinex-centredstrategy that included a finely tuned parameter. The particle Swarms Optimization algorithm improved the parameter, and the input image was transformed to the Hue, Saturation, and Intensity (HSI) model for colour compensation. In addition, by updating multi-scale local information and developing bilateral filtering methods for overcoming artifacts and retaining edges in the dehazing process, the image's overall quality can be improved. To obtain acceptable performance, the computational time, the optimization algorithm convergence velocity, and the edge details information must all be improved.

Deepa Nair et al. [31] suggested a low pass filter for obtaining the transmission map in the RGB, HSV, and Lab three colour spaces as an alternative estimate. The fusion process is used to enhance the naturalness of the output image. When compared to simple convolution, the execution time is much smaller. The centred-surrounded filter improved the speed of transmission depth estimation as well as the memory requirements for the output image. This is beneficial for real-time applications. The proposed image quality assessment methods compare favourably to existing methods, but fine information in the output image was lost.

Dilbag Singh et al. [32] developed Moore neighbourhood-based gradient function to evaluate the effective transmittance and ambient light. To fine-tune the transmittance, GADILF is used. By improving the restoration model, the effect of saturation of pixels and colour distortion from restored images was minimized. There are no halo artifacts, edges, or colour distortions in the output image. According to the performance analysis, dehazed output is restored particularly at the ends of abrupt variations in the obtained depth map. The computational complexity was higher, and texture detail knowledge was lacking.

Table II illustrates the various other image enhancement-based methods concerning the advantages and limitations.

TABLE II. IMAGE ENHANCEMENT BASED DEHAZING METHODS

Author(s)	Technique(s)	Advantage(s)	Limitation(s)
Uche *et al.* [33]	Dynamic Stochastic Resonance (DSR) process, enhancement and wavelet fusion, multi-scale filtering	In terms of execution time, visual perception, and quantitative metrics, algorithm performs better	Hard to implement for high illumination images
Wang *et al.* [34]	Frequency perception assumption	Much higher restored visibility and restored contrast ratios	Colour saturation is visible, but there is some fog observed, and the relationship between the three frequency components is unknown
Vijay Kumar *et al.* [35]	Window-based Integrated Means Filtering (WIMF) with gradient sensitive loss	With reduced halo and gradient artifacts, texture information is greatly preserved and less computational time	Multiple parameter tuning is not considered, not suitable for nighttime hazy images
Cheng-Jian *et al.* [36]	Interval Type-2 Recurrent Fuzzy Cerebellar Model Articulation Controller (IT2RFCMAC) with bilateral filter	The output consists of high visibility details and visible colour contrast ratios	More computing time, as well as the use of special hardware and image transmission compression techniques to be considered
Weiping *et al.* [37]	local image property analysis and linear strength transformation for image enhancement	The dehazed image's luminance, texture, and contrast details are all properly preserved	The thick haze and the vast white background are not taken into account during the dehazing process
Jinxia *et al.* [38]	Unified variational Retinex model	Improved scene radiance recovery and transmission map estimation	Hyper parameters cannot be adjusted adaptively, and computational time is very high
Yang *et al.* [39]	Convolutional networks and wavelet transform with chromatic adaptation	The output image keeps its sharp edges and well-balanced colour data	There have been observed reports of image artifacts and white-region over saturation

C. *Image Fusion Based Dehazing Methods*

Galdran et al. [40] introduced a haze removing approach depending on multi-scale fusion of artificially exposing the original hazy image with gamma corrections while taking into account visual features such as saturation and contrast from the input image. Over-saturated regions can be distinguished from underexposed regions using this approach. The resulting collection of the multiple-exposure images was combined to a haze-free outcome using multiple-scale Laplacians blended approach. In the output image, increased visibility is achieved without any colour shift or compression artifacts, as seen in the DCP model.

Zhiqin et al. [41] created a multi-scale exposure image fusion method with gamma correction, by the analysis of local and global image information using pixel-wise weight map estimation. A multi-scale image fusion technique is used to combine local and global decomposed components based on weight maps. The optimal output image can be retrieved by matching the colour saturation and image luminance. The dehazed method reduced the spatial dependence of the fused image's luminance while balancing its colour saturation. For the best solution of performance using an image fusion technique, factors like exposure time, input image resolution, and imaging system frame rate should be investigated.

Yijing Su et al. [42] devised a framework for estimating global atmospheric light using a novel sky segmentation approach and optimizing the transmission map using a gradient optimization method. A visual sensory selector is used to adaptively choose the images with different exposures in an image fusion strategy based on scale-invariant feature transforms. The second layer of the visual sensory selector is used to choose the most appropriate images. The approach's efficiency can be enhanced by optimizing the execution time with features for each selective layer.

Table III illustrates the various other image fusion-based methods concerning the advantages and limitations.

TABLE III. IMAGE FUSION BASED DEHAZING METHODS

Author(s)	Technique(s)	Advantage(s)	Limitation(s)
Mingyao et al. [43]	Spatial domain-centred saturation and contrast enhancement approach, and image fusion process-based decomposition	Maintains the texture and color details accurately	With image saturation adjustment and block size selection, the method's computational complexity and cost rises
Kumar et al. [44]	Multispectral transmittance fusion	Suitable for hardware implementation, no requirement of any external memory, most likely arbitrary access memory	Cannot be implemented for typical foggy images
Zhao et al. [45]	Patch-wise and pixel-wise transmittance estimation, Multiscale Optimum image fusion (MOF) model	Increased robustness and speed of execution	To minimize the misinterpretation of image colour and depth, an analysis of patch size contributed by the number of scales should be
Haibo et al. [46]	Retinex concept's combination of DCP and Gaussian convolution and image fusion	Better transmittance estimation and use of Morphological operations	Balance between parameters of Morphological operation and thresholding value to obtain the bright areas in the image should be maintained
Shen et al. [47]	Distance-chromaticity correlation and inverse correlation	Improved visibility under bad weather without amplification of noise	Incorrect dehazed output

D. Image Restoration Based Dehazing Methods

Hong et al. [48] created a fusion-based haze removal algorithm to estimate the scene's transmission map from periodic latent stages of scene radiance with defined transmittance values. The image will not be dehazed in its entirety, but the part linked to a particular transmittance value will be retrieved. To obtain latent images useful for image fusion, the same technique will be used for different transmittance values. The image fusion algorithm, which selectively uses patch information, was used to combine the latent images into a haze-free output.

Guoling et al. [49] developed a brightness map based on outdoor hazy scenes to provide both provide brightness reflection and light reflection capability. The transmission map can be calculated using the image fusion strategy based on the estimated brightness map and optimized using the multi-scale guided filter. The contrast of the output image can be increased to a sufficient degree, and the halo artifacts in the scenes of vivid change can be greatly reduced. Time complexity and colour saturation are observed and need to be improved to achieve the algorithm's optimal behaviour.

Wang et al. [50] implemented the Retinex-based colour restoration algorithm and calculated the transmission map at various scales. To enhance adaptively different images, a balance between colour levels, varying dynamic range, and edge information is developed. The dark channel, threshold-based decision image, and hazy image are used to evaluate ambient light. It's hard to get perfect output images without saturation problems by manually selecting the gain factor in the algorithm. Images with thick mist and fog are not properly processed, and chromatic aberration and chronic exposure in bright areas of the image are observed.

Singh et al. [51] developed a DCP for dehazing based on an advanced restoration model. To reduce the chromatic aberration problem in the dehazed picture, the developed model fine-tuned the transmission map. When the joint trilateral filter is used, the projected ambient light is increased. The proposed model has a significant improvement in terms of artifacts introduced according to the qualitative and quantitative analysis and image data loss is observed.

Table IV illustrates the various other image restoration-based methods concerning the advantages and limitations.

TABLE IV. IMAGE RESTORATION BASED DEHAZING METHODS

Author(s)	Technique(s)	Advantage(s)	Limitation(s)
Mingye et al. [52]	Gamma Correction Prior (GCP), quadtree subdivision technique	Recovers accurate object colours and fine target information, less processing time, no aliasing artifacts or saturation issues are observed	The atmospheric scattering was calculated using the depth ratio, which resulted in low accuracy
Ancuti *et al.* [53]	Image matching with local characteristics	Restoration of fine details and gradients from degradation to their full level	Halo artifacts have been added
Singh *et al.* [54]	Fourth-order partial differential equations using trilateral filter (FPDETF)	Fast computation, gradient reversal irregularities and colour distortion have been reduced to their absolute minimum	Colour saturation can be introduced if the patch size is increased
Yibo *et al.* [55]	Estimation of transmission maps using Markov random fields	Colour transmittance close to light is prevented, and blocking irregularities are minimized	Over-saturation of bright regions
Faming *et al.*[56]	Windowing technique to estimate transmission maps and transform them into depth maps	The process is effective in terms of execution time and memory requirements	Irregularities exist, and the issue with the sky area is still present
Wenzong Wan *et al.* [57]	The DCP and Markov Random Field hybrid model (MRF)	No requirement of refinement filter, the misclassified transmittance is corrected, and the anti-mist effect is improved	To minimize the computational cost and blocking objects, the number of labels is to be manually selected

E. Image Segmentation Based Dehazing Methods

Salazar-Colores et al. [58] devised a method for dehazing based on DCP, which included determining the transmittance from a minimal channel, morphological operations, and Gaussian filtering. The benefit of this algorithm is that it has a faster execution time because it uses morphological processing. The resulting performance was addicted to noise, most likely speckle noise and impulse noise, lowering the image's dominance.

Liu et al. [59] demonstrated a dehazing technique focused on extended sky area segmentation and a Multiscale Opening Dark Channel Model (MODCM). The method of estimating air-light begins with the identification of large-scale sky regions using SVM categorization. Two separate novel techniques for obtaining precise atmospheric light are applied based on sky region classification. Furthermore, different block sizes for different edge levels can be used to customize the dark channel model's halo artifacts. To preserve edge information and to fine-tune the transmission map, guided filtering is used. The approach works by classifying sky regions and applying different strategies, resulting in a better haze-free image with colour authenticity. The disadvantages are computation time and the inability to use colour and texture knowledge for superior classification.

Hassan et al. [60] suggested using the super-pixel segmentation approach to estimate white light. The hazy region is selected using the maximum super-pixel intensity, and atmospheric light is defined as the highest super-pixel intensity among the three channels. Following that, a rolling directed filter is used to determine and refine the transmittance. This type of filter preserves scene details in the output image, most likely in the form of structures, edges, and textures. Because of the fast processing speed of high-dimensional images, the proposed method has many real-time applications. To minimize computational complexity, execution time, and the method's applicability, hierarchal-based deep learning models have to be designed.

Hsueh-Yi et al. [61] used the fuzzy inference model to estimate the ambient light variations, a commonly used imaging technique for segmentation, filtering, reconstruction, and enhancement. Basic morphological processing operations such as erosion and Back Propagation neural network layers can be used to refine transmission map and to remove irregularities. Scene radiance and colour contrast are not realized in the fuzzy system with fewer fuzzy layers, and noise remains in the output image with more fuzzy layers.

Trung Minh et al. [62] suggested a fuzzy-based colour ellipsoid model appropriate for RGB pixel clusters to estimate the transmittance depth from the mean value of the hazy pixels. The high-level refinement method is eliminated in this model, which allows for simultaneous estimation and refinement of transmittance. To perform segmentation, the fuzzy approach compares pixel positions and differences in pixel value variations. The prior vector's position in the same patch centred pixel is used to measure the threshold value. The model takes less time to execute and improves contrast without irregularities.

F. Deep Learning Based Dehazing Methods

Fig. 2 depicts deep learning-centred image dehazing, which consists of two approaches, one based on supervised learning and the other on unsupervised learning. Supervised learning gets trained by the haze-free images and generates the output; whereas unsupervised learning doesn't get trained rather it generates an output centred on input image features.

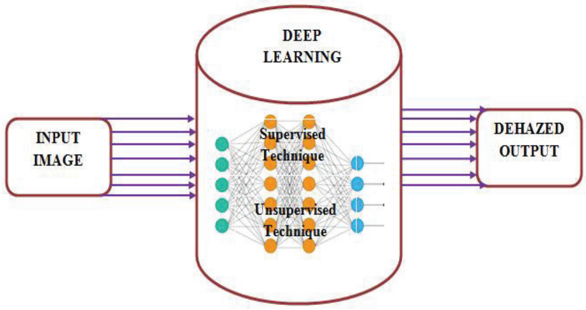

Fig. 2. Block diagram representing deep learning-based image dehazing

Jinjiang et al. [63] introduced a dehazing technique with two stages and focused on residual-centred deep CNN: Initially, the transmittance depth was determined by optimizing the loss function among the corresponding ground truth and reconstruction transmission using the residual learning network; later, the ratio of the hazy image to transmittance depth was considered as input. There was no image blur or colour distortion observed by the algorithm. The need for large datasets for training to boost the efficiency of the Residual-centred deep CNN is a difficult aspect to overcome.

Song et al. [64] proposed to add a novel Ranking layer to the traditional CNN so that the same time both the statistical and structural properties of hazy input could be calculated. The network was used to automatically extract dominant haze-relevant features. Transmission depth can be calculated using a haze density prediction model trained using random forest regression. Although the network performs well and produces better results, the vanishing gradient descent issue is caused by this ranking CNN.

Zhihui et al. [65] devised a quadtree decomposition method to precisely measures airlight light and eliminate the white region's effect on estimation. To analyze the input image, three convolution layers were used, as well as feature learning, the propagation map was measured and fine-tuned, and image fusion techniques were used to avoid the shortcomings of physical extraction of features. The method has a high-quality impact on pattern information, and it is extremely robust for any dataset. The method's key flaw is its irrelevant aspect, which contributes to inaccurate haze removal.

Zheng et al. [66] developed a standard and agnostic CNN that wasn't designed explicitly for image dehazing and had no knowledge of the physical scattering model. The network is made up of local and global residual learning stages, which fully eliminates the need for individual parameter estimation.

Ren et al. [67] introduced a multi-scale based deep neural network that learns the association between input and its corresponding scene depth. A coarse-scale network and fine-scale network predicts a global scene depth, and refinement of local dehazed outputs in the algorithm. A global edge-filter network was also designed to fine-tune the approximate scene depth edges. The algorithm performs well in terms of superiority and execution time, but it takes a longer time to learn.

Haouassi et al. [68] devised a system for accurate airlight estimation based on haze image blur and quadtree decomposition, as well as a cascaded multi-scale convolutional architecture for transmittance estimation and fine-tuning. The architecture's sub-network was made up of three-layer Dense-units. The method accurately removes haze from night time hazy input images while retaining colour levels similar to haze-free images. However, false regions were discovered, lowering the method's accuracy.

Table V illustrates the various other image Deep Learning-based methods concerning the advantages and limitations.

TABLE V. DEEP LEARNING BASED DEHAZING METHODS

Author(s)	Technique(s)	Advantage(s)	Limitation(s)
Zhang et al. [69]	Deep Residual Convolutional Dehazing Network (DRCDN)	Faster execution time, no irregularities observed in output	More complicated, resulting in unbalanced data training, which reduces the technique's efficiency
Hua et al. [70]	Iterative residual network with residual blocks inside the computing unit	Effective in removing haze from the image	Manual selection of the number of iterations in the concatenating device made the method less accurate
Kailei et al. [71]	Conditional dependent Generative Adversarial Networks	Improved atmospheric model with efficiency in dehazing	Difficult to design and time-consuming
Yan Zhao *et al.* [72]	Guided Conditional Generative Adversial Network with Multiscale discriminator	Exceptional results for both indoor and outdoor dataset images	Concatenating previous data with different descriptions can result in inaccuracy
Dhara *et al.* [73]	Weighted least squares filtering	There are no issues with saturation or colour loss, and the colour cast has been removed	A high rate of false positives and a poor precision value
Chen *et al.* [74]	Enhanced DCP with Cycle Generative Adversarial Network	Effective processing of high haze images, as well as mist images of great importance in complex scenes	Reduced representation capability due to the model's over complexity and computational inefficiency
Zhang *et al.* [75]	Local Multi-scale Hierarchical Prediction Method and Generative Adversarial Dehazed Mapping Nets (GADMN)	To incorporate simulated intensity and disperse reflected light that appeared in mist, it used the Multi-light scattered technique	It suffered from ill-posed nature
Dacheng Tao *et al.* [76]	Encoder network with fusion model for learning the haze-free image	Computationally effective and lightweight, with high restoration accuracy	Colour artifacts appear, as well as incorrect transmission in sky regions and the inability to effectively eliminate heterogeneous haze
Tiantong Guo et al. [77]	EDN-3J, EDN-AT, and EDN-EDU are used in an ensemble technique	Reduced variance in implementation with sub-models in the case of non-homogeneous haze	The sub-model is less capable due to its limited structure, and the requisite number is also very large for significant change and avoiding heavy architecture
Yizhou *et al.* [78]	Physical disentanglement based network using an unsupervised approach	Scene depth information is required, colour recovery is more	The light rays emitted from the target surface are scattered by aerosol particles, resulting in light intensity attenuation
Dudhane *et al.* [79]	Generative Adversarial Network with skip connections	Colour distortion was minimized to a greater degree, and visibility in the presence of aerosol particles was increased	Difficult to manipulate outdoor images with georeferenced digital terrain and urban environments
Suárez *et al.* [80]	Dense convergence of network for cross-spectral images	Multiple loss function scheme, shorter connections in learning layers, texture loss improvement, feature re-use	For improved generalization, ground truth haze-free images are needed, as well as a number of loss functions to enhance the learning process

III. PERFORMANCE EVALUATION

When compared to other dehazing techniques in terms of performance, the techniques GADILF [28], IT2RFCMAC [36], MOF [45], MRF-DCP [57], MODCM [59], DRCDN [69], and GADMN [75] are found to be better. Techniques are evaluated based on performance measures such as Sensitivity, Precision, Accuracy, and Specificity, and Fig. 3 shows a graphical representation of the performance of the technique.

When compared to other methods, the DRCDN and GADMN methods achieve 88.96%and 89.99% accuracy, respectively. The GADILF and IT2RFCMAC achieve the lowest performance metrics range of 75.45% to 81.25%, while the techniques MOF, MRF-DCP, and MODCM achieve a metrics value range of 76.85% to 86.89%. When compared to DRCDN, which achieves 82.35% and 82.5%, the GADMN has a lower sensitivity and precision value of 81.52% and 81.7%.

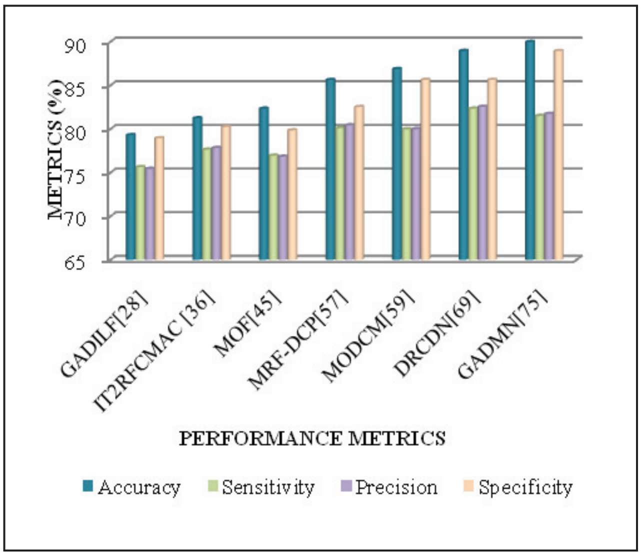

Fig. 3. Analysis of Various Dehazing Technique based on Performance Metrics

IV. Conclusion

Image dehazing is a difficult problem that is caused by the uneven and non-uniform distribution of haze. This causes inaccuracy in computer vision, change analysis applications etc., An empirical review on various dehazing techniques based on Single image, Multiple image, Image Enhancement, Fusion, Restoration, Segmentation, Deep Learning-centred techniques with the advantages, limitations is presented and corresponding performance is evaluated. Continuous development and improvement in techniques with respect to different problems should be made. It could be inferred based on the review that the continuous learning centric techniques give a better performance when analyzed with the other methods, and it could be employed in LC detection. In Deep Learning, accuracy can be attained by the CNN and GAN, however, the main shortcoming is that they utilize additional time for training, and the computational complexity is more. Hence, it is necessary to model an extremely consistent and intelligent optimized Deep Learning based dehazing technique suitable for real time applications. For maintaining the model's efficacy, the designed model must be competent in handling uncertainties along with several artifacts.

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